

II. SECURITY PRINCIPLES

Aim:

Learn about the security triad and common security models and principles.

Procedure:

1. Study the CIA Triad (Confidentiality, Integrity, Availability).
2. Review core principles like least privilege, defense in depth, and zero trust.
3. Understand basic security models such as Bell-LaPadula Model and Biba Model.

TASKS:

Task 1: Introduction

Think how you would describe something as secure.

No answer needed

✓ Correct Answer

Task 2: CIA

Click on "View Site" and answer the five questions. What is the flag that you obtained at the end?

THM{CIA_TRIAD}

✓ Correct Answer



Task 3: DAD

The attacker managed to gain access to customer records and dumped them online. What is this attack?

Disclosure

✓ Correct Answer

A group of attackers were able to locate both the main and the backup power supply systems and switch them off. As a result, the whole network was shut down. What is this attack?

Destruction/Denial

✓ Correct Answer

Task 4: Fundamental Concepts of Security Models

Click on "View Site" and answer the four questions. What is the flag that you obtained at the end?

THM{SECURITY_MODELS}

✓ Correct Answer

Task 5: Defence-In-Depth

Make sure you have read the above.

No answer needed

✓ Correct Answer

Task 6: ISO/IEC 19249

Which principle are you applying when you turn off an insecure server that is not critical to the business?

2

✓ Correct Answer

Your company hired a new sales representative. Which principle are they applying when they tell you to give them access only to the company products and prices?

1

✓ Correct Answer

While reading the code of an ATM, you noticed a huge chunk of code to handle unexpected situations such as network disconnection and power failure. Which principle are they applying?

5

✓ Correct Answer

Task 7: Zero Trust Versus But Verify

Make sure you have read the above.

No answer needed

✓ Correct Answer

Task 8: Threat Versus Risk

Make sure you have read the above.

No answer needed

✓ Correct Answer

Task 9: Conclusion

Make sure you have taken notes of all the key terms and acronyms we covered in this room.

No answer needed

✓ Correct Answer

Result:

Gained understanding of the CIA triad and its role in security. Learned key principles like least privilege and layered defense. Developed basic knowledge of common security models used to enforce confidentiality and integrity.